

KENYA NATIONAL EXAMINATIONS COUNCIL

KCPE 2014

SCIENCE

3850724

READ THESE INSTRUCTIONS CAREFULLY

Time:1 hour 40 minutes

1. You have been given this question booklet and a separate answer sheet. The question booklet contains 50 question
2. Do any necessary rough work in this booklet.
3. When you have chosen your answer, mark it on the **ANSWER SHEET**, not in this question booklet.
4. Use only an ordinary pencil.
5. Make sure that you have written on the answer sheet:

YOUR INDEX NUMBER

YOUR NAME

NAME OF YOUR SCHOOL

6. By drawing a dark line inside the correct numbered boxes mark your full Index Number(i.e School code number and the three-figured Candidate's number)in the grid near the top of the answer sheet.
7. Do not make any marks outside the boxes.
8. Keep the sheet as clean as possible and do not fold it.
9. For each of the question 1-50 four answers are given. The answers are lettered A,B,C and D. In each case only ONE of the four answers is correct. Choose the correct answer.
10. On the answer sheet the correct answer is to be shown by drawing a dark line inside the box in which the letter have chosen is written.

Example

In the Question booklet.

4. Which one of the following components of blood is involved in the clotting after injury?

- A. Plasma
- B. White blood cells
- C. Red blood cells
- D. Platelets

The correct answer is D

1 [A] [B] [C] [D] 14. [A] [B] [C] [D] 24. [A] [B] [C] [D] 34. [A] [B] [C] [D]

11. Your dark line MUST be within the box.
12. For each question ONLY ONE box is to be marked in each set of four boxes.
This question paper consists of 7 printed pages and 1 black page.

The Kenya National Examination Council, 2014.

1. In the human body digested food is absorbed in the
 - A. Stomach
 - B. Large intestine
 - C. Rectum
 - D. Small intestine
2. The diagram below represents a certain type of human tooth.
The function of the tooth is
 - A. Gripping and tearing
 - B. Tearing and grinding
 - C. Crushing and grinding
 - D. Biting and crushing
3. A certain person looked healthy and strong but tested positive for HIV/AIDS. In which one of the following stages of HIV/AIDS was the person likely to be?
 - A. Window stage
 - B. Incubation
 - C. Symptomatic
 - D. Full blown
4. The following are sexually transmitted infections **except**
 - A. Syphilis
 - B. Gonorrhea
 - C. Bilharzia
 - D. Chancroid
5. Which of the following pairs consists of planets with the longest orbits?
 - A. Mars and mercury
 - B. Earth and Saturn
 - C. Venus and Jupiter
 - D. Neptune and Uranus
6. In order to make a certain weather instruments, pupil assembled the following materials:
Glass bottle, ink, biro pen tube, nails, Manila paper and water
Which one of the following instruments was the pupil likely to make?
 - A. Wide vane
 - B. Liquid thermometer
 - C. Rain gauge
 - D. Wind sock
7. The following are some adaptations of plants that grow in wet environment **Except** having
 - A. LARGE FLAT LEAVES
 - B. Leaves with wax upper surface
 - C. Thick leaf cuticle
 - D. An increased number of stomata
8. Which one of the following groups consists **only** of field pests?
 - A. Weevils,aphids and weaver birds
 - B. Aphids ,weaver birds and white ants
 - C. Weevils ,white ants and stock borer
 - D. Aphids , weaver birds and stock borer
9. Which one of the following is a characteristics of amphibians?
 - A. Lay their eggs on land
 - B. Live party in water
 - C. Have most skin with scales
 - D. Adults breathe through the gills
10. The diagram below represents a beak of a certain bird.
The bird represented is a
 - A. Filter feeder
 - B. Grain eater
 - C. Nectar eater
 - D. Fish eater
11. In which one of the following activities is water used in industries?
 - A. Cleaning tools and equipment
 - B. Preparation of soft drinks
 - C. Washing cars
 - D. Watering plants
12. Which one of the following is advantage of hard water?
 - A. Saves soap during washing
 - B. Lathers easily with soap

- C. Forms scum with soap
- D. Suitable for drinking

13. Mary and John were playing on the sea saw as shown in the diagram below.

In order to balance on the sea saw

- A. John should move towards the pivot
- B. Mary should move away from the pivot
- C. Move the pivot towards John
- D. Move the pivot towards Mary

14. The following are methods of maintaining simple tools:

- i) cleaning after use
- ii) Proper storage
- iii) Sharpening sharp edges
- iv) Proper use of tools

Which pair consists only of methods for safety from accident?

- A. (ii) and (iv)
- B. (i) and (iv)
- C. (ii) and (iii)
- D. (i) and (iii)

15. Which none of the following components of Air is used in preservation of soft drinks?

- A. Carbon dioxide
- B. Nitrogen
- C. Oxygen
- D. Inert gases

16. Which one of the following pairs of processes are as a results of increase and decrease in temperature?

Increases
decrease

- A. Freezing
condensation

- B. Expansion
evaporation

- C. Contraction
expansion

- D. Melting
condensation

17. Which one of the following statement is true about liquids and gases?

- A. Liquid have definite volume
- B. Gases have definite volume
- C. Liquid have definite shape
- D. Gases have definite shape

18. A mixture of salt solution, maize grains and iron filings can be separated by

- A. Filtering, use of magnet, evaporation
- B. Decanting ,picking, winnowing
- C. Filtering ,decanting, evaporation
- D. Winnowing, use of magnet ,picking

19. The energy transformation that takes place when a touch that uses batteries is switched on is

- A. Electrical ->chemical->heat->light
- B. Electrical-> heat ->chemical->light
- C. Chemical->electrical->light ->heat
- D. Chemical->electrical->heat->light

20. Formation of a rainbow can be demonstrated by the following activities except when

- A. Observing the sun through a horizontally held transparent biro pen
 - B. Placing mirror in water to face the sun
 - C. Spraying water in the air from the mouth in the bright sunshine
 - D. Placing biro pen casing on water in a glass container
21. Which one of the following is a source of electricity?
- A. Touch
 - B. Dam
 - solar panel
 - C. Biogas plant
22. Which one of the following consists of foods that are **mainly** for body building?
- A. BEEF ,CASSAVA, POTATOES
 - B. Mutton ,maize, pawpaw
 - C. Peas, beans, eggs
 - D. Carrots, bananas, tomatoes
23. Which one of the following groups of symptoms is **correctly** matched with deficiency disease?
- Rickets
 - anemia
 - kwashiorkor
 - A. Bowlegged, brown hair, wrinkled face
 - B. Knocked-knees, pale eyes potbellied
 - C. Rapid heartbeat potbellied, breathlessness
 - D. Brown hair wrinkled face, bowlegged

24. Which one of the following pairs consists only of components of environment that produced carbon dioxide
- A. Air and plants
 - B. Animals and plants
 - C. Water and air
 - D. Soil and animals
25. Use of excess fertilizers will mainly pollute
- A. Soil and water
 - B. Water only
 - C. Soil and air
 - D. Soil only
26. Which one of the following components of blood is correct matched with its function?
- | Blood component | function |
|------------------------|-------------------|
| A. Red blood cells | clotting of blood |
| B. White blood cells | transport oxygen |
| C. Platelets | fight germs |
| D. Plasma | transport food |
27. The following are some effects of drug abuse:
- i) Rape
 - ii) Addiction
 - iii) Truancy
 - iv) Impaired judgement
 - v) Drug induced accidents
- Which one of the consists only of social effects
- A. (i) (ii) (iv)
 - B. (ii) (iii) (iv)
 - C. (i) (iii) (iv)
 - D. (i) (iv) (iii)

28. Pupils constructed a weather instrument shown below.

The mistake made in the construction was

- A. Using coloured water
- B. Placement of the scale
- C. Using a narrow tube
- D. Closing the cork tightly

29. Which one of the following pairs consists only of plants that directly feed on insects

- A. Pitcher plant and sundew
- B. Mushroom and butterwort
- C. Sunflower and pitcher plant
- D. Sundew and toadstool

30. The chart below shows a simple classification of plants.

Which one of the following pairs represents Y and Z?

Y

Z

- A. Carrot
sisal
- B. Kale
onion
- C. Moss
cypress
- D. Sugarcane
fern

31. Which one of the following pairs consists only farm animals that produce mutton?

- A. Cow and goat
- B. Goat and sheep
- C. Cow and sheep
- D. Sheep and pig

32. The following are some characteristics of animals:

- i) Varying body temperature
- ii) Lay eggs
- iii) Have scales
- iv) Have back born

Which two characteristics are for both toad and duck?

- A. (i) and (iii)
- B. (i) and (ii)
- C. (ii) and (iv)
- D. (iii) and (iv)

33. The method of grazing that would require the largest piece of land to practice is

- A. Herding
- B. Stall feeding
- C. Tethering
- D. Paddocking

34. Which of the following pairs of machines consists of only of inclined plans

- A. Larder and staircase
- B. Staircase and wheelbarrow
- C. Claw hammer and ladder
- D. Spade and crew ladder

35. In which one of the following is friction an advantage?

- A. Skating on ice
- B. Swimming in water
- C. Production of heat in moving parts
- D. Braking of bicycle

36. In an investigation of factors affecting sinking and floating pupils carried out the following activities

- | | |
|--|--|
| <p>i) Placed a bottle top in water</p> <p>ii) Made the same bottle top into a ball and placed it on water</p> <p>Which one of the following factors were the pupils investigating?</p> <p>A. Size</p> <p>B. Shape</p> <p>C. Type of materials</p> <p>D. Mass</p> <p>37. Heat transfer in liquids and gases mainly takes place through</p> <p>A. Radiation</p> <p>B. Convection and conduction</p> <p>C. Convection</p> <p>D. Conduction and radiation</p> <p>38. Which one of the following is a transparent materials</p> <p>A. Kerosene</p> <p>B. Mirror</p> <p>C. Frosted glass</p> <p>D. White paper</p> <p>39. Which one of the following is safety measure against lighting when is raining?</p> <p>A. Leaning against walls</p> <p>B. Walking on open areas</p> <p>C. Wearing rubbers shoes</p> <p>D. Using an umbrella</p> <p>40. Which one of the following nutrients is required for formation of blood in expectant mothers?</p> <p>A. Vitamin D</p> <p>B. Calcium</p> <p>C. Phosphorous</p> <p>D. iron</p> | <p>41. Which one of the following soil conservation measures is least suitable on a large piece of land?</p> <p>A. Terracing</p> <p>B. Mulching</p> <p>C. Trees planting</p> <p>D. Contour farming</p> <p>42. When pupils heated some garden soil, smoke was produced. The observation made indicated the presence of</p> <p>A. Organic matter</p> <p>B. Air</p> <p>C. Mineral particles</p> <p>D. water</p> <p>43. The following are some characteristics of soil:</p> <p>i) Fine particles</p> <p>ii) Cracks when dry</p> <p>iii) Poor drainage</p> <p>iv) Good capillarity</p> <p>The characteristics are for</p> <p>A. Loam soil</p> <p>B. Loam and sand soil</p> <p>C. Clay soil only</p> <p>D. Loam and clay soil</p> <p>44. Pupils went for a nature walk and observe a plant with yellow fruits and thorns on the stem. The plant observed was likely to be</p> <p>A. Mexican marigold</p> <p>B. Pig weed</p> <p>C. Black jack</p> <p>D. Sodom apple</p> <p>45. Which one of the following groups only consists of materials that are non-magnetic?</p> |
|--|--|

- A. Piece of glass, silver coin, cello tape
 - B. Nail, steel wool, spoon
 - C. Scissors, razor blade, piece of glass
 - D. Staple pins, sowing needles, aluminum
46. Which one of the following groups consists only gases that comprises 0.03, 0.97, and 21 percentage in air respectively?
- A. Inert gases, oxygen and nitrogen
 - B. Carbon dioxide, inert gases and oxygen
 - C. Inert gases carbon dioxide and nitrogen
 - D. Carbon dioxide, oxygen and nitrogen
47. Which one of the following is not a component of soil?
- A. Rock particles
 - B. Water
 - C. Plants
 - D. Air
48. The diagram below represents a set up that can be used to investigate good and poor conductors of electricity?

Which one of the following materials when used to connect wires point S would make the bulb not light?

- A. Pencil
 - B. Aluminum foil
 - C. Steel wool
 - D. Rubber band
49. The diagram below represents a claw hammer in use

Which pair of the positions V, W, X and Y represents the load and pivot?

- A. V and Y
 - B. W and x
 - C. V and X
 - D. W and Y
50. In a certain investigation pupils sat at different position around a pupils who was ringing a bell. Which one of the following aspects of sound were the pupils investigate?
- A. SPECIAL SOUNDS
 - B. Loud and softy sound
 - C. Noise pollution
 - D. Direction of sound